Universal Basic Income in the Age of Automation: A Critical Exploration and Policy Framework

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16 May 2025

Abstract

The increasing sophistication of artificial intelligence (AI) and automation technologies has precipitated significant debate regarding their potential to cause widespread job displacement. This paper critically examines the proposition that Universal Basic Income (UBI) or other forms of guaranteed basic income (GBI) could serve as a comprehensive solution to the challenges posed by such technological unemployment. Through a synthesis of existing empirical evidence from BI/GBI pilot programs, qualitative analyses of automation's impact on labour markets, and quantitative data, a nuanced perspective is presented. It is argued that while basic income demonstrates considerable promise in alleviating poverty, enhancing economic security, and improving certain well-being outcomes for affected individuals, its efficacy as a singular panacea is limited. The analysis considers the dual nature of automation, which involves both job displacement and job transformation. Key findings from BI experiments related to labour supply, health, and education are weighed against concerns regarding economic feasibility, potential inflationary pressures, and the non-monetary aspects of work. Common themes, ongoing debates, and critical research gaps in the current literature are identified. The paper concludes that basic income is more appropriately viewed as a potentially crucial component within a broader, multifaceted policy framework designed to navigate the future of work. This framework should also encompass robust investments in education and skills development, active labour market policies, and strategies to ensure the equitable distribution of productivity gains from automation.

Keywords: Universal Basic Income, Automation, Artificial Intelligence, Job Displacement, Technological Unemployment, Labour Markets, Social Policy, Economic Security.

Citation Suggestion: Tan, K. H. (2025). *Universal Basic Income in the Age of Automation: A Critical Exploration and Policy Framework*. Available at: https://www.researchgate.net/profile/Kwan-Hong-Tan/publications

Author's Note: Dr Tan Kwan Hong serves in professorship, lecturing, mentoring and supervisory capacities at multiple global universities, including the Singapore University of Social Sciences, the University of West London (UK), the University of Suffolk (UK), the University of Roehampton (UK), TETR College of Business (USA), European International University (France), Graham International University (USA), the European Institute of Management and Technology (Switzerland), Central Global University (Georgia), Hanbridge Institute (Singapore), Aventis Graduate School (Singapore), and the At-Sunrice GlobalChef Academy (Singapore).

Introduction: The Spectre of Technological Unemployment and the Promise of Basic Income

The relentless advancement of artificial intelligence (AI) and automation technologies is posited to precipitate significant transformations in labour markets globally. Projections of widespread job displacement across various sectors have fuelled a robust academic and public discourse concerning the societal and economic ramifications (Subaveerapandiyan & Shimray, 2024; Georgieff & Hyee, 2022). While some scholarship emphasizes the complementary nature of AI and human labour, potentially leading to new job creation and productivity gains (Wang & Lu, 2025; Autor, 2015, as cited by Wang & Lu, 2025), considerable concern remains regarding the vulnerability of specific occupations and skill sets, potentially exacerbating economic insecurity and inequality (Frey & Osborne, 2017, as cited by Wang & Lu, 2025).

In this context, Universal Basic Income (UBI), or more broadly, Guaranteed Basic Income (GBI), has been advanced as a prominent policy proposal. It is conceptualized as a mechanism to provide a foundational level of economic security, thereby mitigating the adverse effects of technological unemployment, alleviating poverty, and potentially fostering individual well-being and societal stability. The conceptual relationship, as illustrated in Figure 1 (see Appendix A), depicts AI and automation leading to job displacement, which in turn can result in economic insecurity and reduced well-being. UBI is presented as an intervention aimed at counteracting these negative outcomes by promoting poverty reduction, enhancing well-being, and ensuring economic stability.

This paper critically examines the question: **Could basic income be the answer to jobs lost to automation and AI?** The argument is advanced, through a synthesis of empirical evidence from BI/GBI pilot programs (Rizvi et al., 2024; Hasdell, 2020; University of Chicago, Inclusive Economy Lab, 2024), qualitative analyses of automation impacts, and quantitative data on labour market trends, that while basic income presents a compelling partial solution with demonstrable benefits in certain domains, it is unlikely to be a panacea for the multifaceted challenges posed by large-scale, automation-induced job displacement. A passive voice is maintained throughout this critical academic exploration.

The Dual Nature of Automation: Displacement and Transformation

The impact of automation and AI on employment is not monolithic. Research indicates a dual effect: displacement of labour in routine or automatable tasks, and transformation of existing jobs alongside the creation of new roles that complement these technologies (Wang & Lu, 2025). The net effect on aggregate employment remains a subject of intense debate and ongoing research. However, the distributional consequences are a significant concern; even if new jobs are created, they may require different skills, be located in different geographical areas, or offer different levels of remuneration and security than those lost. This skills gap and transitional friction can lead to periods of unemployment or underemployment for affected workers.
Basic Income: Empirical Evidence and Theoretical Underpinnings

Numerous basic income experiments and pilot programs have been conducted globally, providing valuable, albeit context-specific, empirical data. Systematic reviews (Rizvi et al., 2024; Hasdell, 2020) suggest several key findings:

- **Poverty and Well-being:** A consistent finding across many studies is the positive impact of GBI on poverty reduction, particularly concerning food security. Improvements in subjective financial well-being, life satisfaction, and mental health outcomes are also frequently reported. These findings lend credence to the argument that BI can provide a crucial safety net, alleviating the immediate economic distress associated with job loss.

- **Labour Supply:** The impact on labour supply is one of the most contentious aspects. Evidence is mixed: some studies report minimal to no significant reduction in work participation, while others indicate modest declines, often concentrated among secondary earners, individuals pursuing education, or those with caregiving responsibilities (Rizvi et al., 2024). The concern that BI would lead to widespread withdrawal from the labour force is not strongly supported by most pilot program data, though the scale and duration of these experiments may not fully capture long-term societal adjustments.

- **Health and Education:** Some studies indicate positive impacts on health outcomes and educational attainment, potentially due to reduced stress and increased financial capacity to invest in human capital.

Critical Evaluation: Basic Income as The Solution

While the evidence suggests BI can be an effective tool for poverty alleviation and enhancing well-being, its capacity to serve as the primary or sole answer to automation-induced job loss is questionable for several reasons, as identified in the prior analysis (Appendix B and Appendix C are provided as supplementary materials):

- **Scale and Economic Feasibility:** Implementing a truly universal basic income at a level sufficient to replace lost wages for a significant portion of the population would entail substantial fiscal costs. The political and economic challenges of funding such a program through taxation or reallocation of resources are immense and largely unresolved.

- **Addressing Non-Monetary Aspects of Work:** Employment provides not only income but also social engagement, identity, and a sense of purpose for many individuals. While BI can replace lost income, it does not inherently address these non-monetary losses. The societal implications of a large segment of the population being detached from traditional work structures, even if economically supported, require careful consideration.

- **Inflationary Pressures:** A large-scale injection of demand through UBI, if not matched by corresponding increases in productivity and supply, could lead to inflationary pressures, potentially eroding the real value of the basic income itself.

- **The Evolving Nature of Work:** Automation is not simply about job destruction but also job transformation. A policy response focused solely on passive income support might neglect the crucial need for investment in reskilling, upskilling, and lifelong learning initiatives to help workers adapt to the changing demands of the labour market.

BI could, however, provide the necessary security for individuals to undertake such retraining.

- **Heterogeneity of Impacts and Needs:** The effects of automation and the needs of displaced workers are diverse. A uniform BI payment might not be the most efficient or equitable solution for all individuals and circumstances. Tailored support systems, potentially including BI as one component, might be more appropriate.

The Path Forward: Basic Income within a Broader Policy Framework

It is argued that basic income should not be viewed as a standalone silver bullet but rather as a potentially vital component of a comprehensive policy strategy to navigate the future of work in an era of advanced automation. Such a strategy would need to encompass:

- **Investment in Education and Skills Development:** Robust programs for reskilling, upskilling, and adapting educational systems to foster skills complementary to AI and automation.

- **Active Labour Market Policies:** Support for job search, mobility, and entrepreneurship.

- **Social Safety Net Modernization:** Integrating BI with existing welfare programs to create a more streamlined, effective, and non-stigmatizing safety net.

- **Policies to Shape Technological Development:** Encouraging innovation that complements rather than solely displaces human labour, and ensuring that the productivity gains from automation are broadly shared.

- Addressing Regional Disparities: Targeted interventions for communities disproportionately affected by automation.

Conclusion

The question of whether basic income could be the answer to jobs lost to automation and AI is complex and does not yield a simple affirmative or negative response. The evidence strongly suggests that basic income can be a powerful tool for poverty reduction, enhancing economic security, and improving individual well-being, thereby addressing some of the critical negative consequences of technological unemployment. However, concerns regarding labour market incentives, macroeconomic feasibility, and the non-monetary aspects of work suggest that BI, in isolation, is unlikely to be a comprehensive solution. A more nuanced approach is warranted, wherein basic income is considered as a foundational element within a broader, multifaceted policy framework designed to support individuals and communities through the profound economic and social transitions driven by AI and automation. Further research, particularly on long-term impacts, scalability, and integration with other policies in the specific context of automation-driven job displacement, is crucial to fully understand the potential and limitations of basic income in shaping an equitable and prosperous future.

References

Autor, D. H. (2015). Why Are There Still So Many Jobs? The History and Future of Workplace Automation. Journal of Economic Perspectives, 29(3), 3–30.

Frey, C. B., & Osborne, M. A. (2017). The Future of Employment: How Susceptible are Jobs to Computerisation?. Technological Forecasting and Social Change, 114, 254–280.

Georgieff, A., & Hyee, R. (2022). Artificial Intelligence and Employment: New Cross-Country Evidence. Frontiers in Artificial Intelligence, 5, 832736. https://doi.org/10.3389/frai.2022.832736

Hasdell, R. (2020) . What We Know About Universal Basic Income: A Cross-Synthesis of Reviews. Stanford Basic Income Lab. Retrieved from https://basicincome.stanford.edu/uploads/Umbrella%20Review%20BI_final.pdf

Rizvi, A., Kearns, M., Dignam, M., Coates, A., Sharp, M. K., Magwood, O., Labelle, P. R., Elmestekawy, N., Rossiter, S., Al-Zubaidi, A. A. A., Dewidar, O., Idzerda, L., Aguilera, J. M. P., Seal, H., Little, J., Antequera Martín, A. M., Petkovic, J., Jull, J., Gergyek, L., … Kristjansson, E. (2024). Effects of Guaranteed Basic Income Interventions on Poverty-Related Outcomes in High-Income Countries: A Systematic Review and Meta-Analysis. Campbell Systematic Reviews, 20(2), e1414. https://doi.org/10.1002/cl2.1414

Subaveerapandiyan, A., & Shimray, S. R. (2024) . The Evolution of Job Displacement in the Age of AI and Automation: A Bibliometric Review (1984–2024). Open Information Science. Published online December 23, 2024. https://doi.org/10.1515/opis-2024-0010

University of Chicago, Inclusive Economy Lab (Urban Labs) . (2024, September). Cash Rules Everything Around Me: A Summary of Existing Research on Guaranteed Income. Retrieved from https://urbanlabs.uchicago.edu/attachments/5ff88f36218ed7d9bee03d8b2e3f5e87998

b51c9/store/71b4d2b6c98f6d7d0e4b8e5ac62db72a0139a748a1dc59ee192309d9e7cf/092024+IEL+GI+Literature+Review.pdf

Wang, K.-H., & Lu, W.-C. (2025). AI-Induced Job Impact: Complementary or Substitution? Empirical Insights and Sustainable Technology Considerations. *Sustainable Technology and Entrepreneurship*, 4(1), 100085. (Available online 16 February 2024). https://doi.org/10.1016/j.stae.2024.100085

Appendix A: Conceptual Diagram

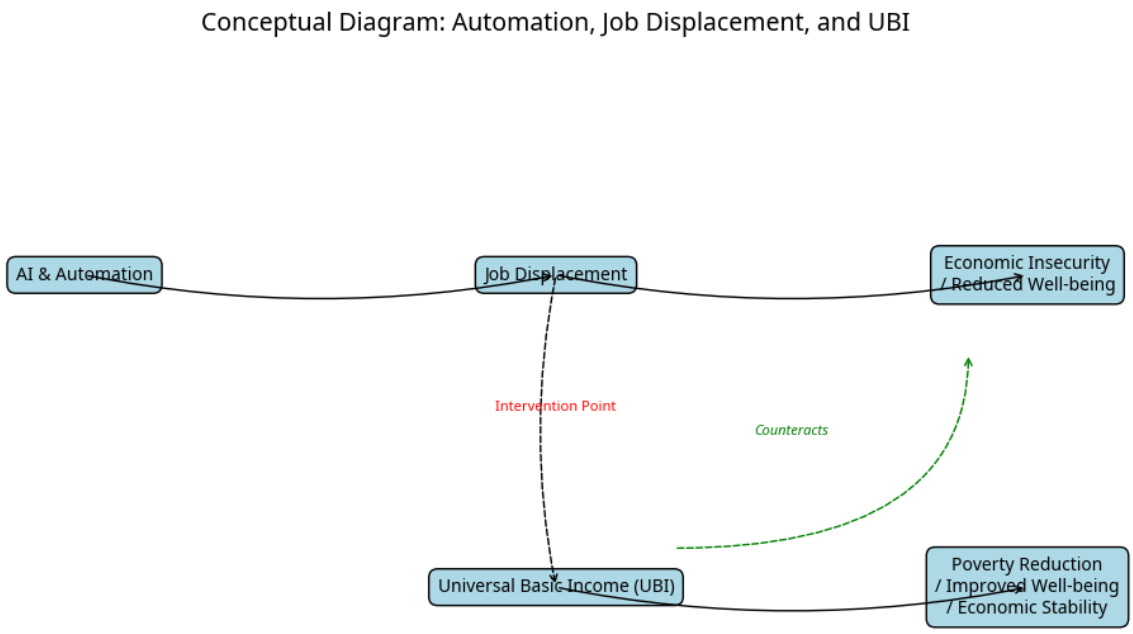

Figure 1: Conceptual Diagram: Automation, Job Displacement, and UBI

This diagram illustrates the posited relationships where AI & Automation contribute to Job Displacement, leading to potential negative consequences such as Economic Insecurity and Reduced Well-being. Universal Basic Income (UBI) is shown as an intervention point stemming from Job Displacement, aiming to counteract these negative effects by fostering Poverty Reduction, Improved Well-being, and Economic Stability. A dashed green arrow also indicates UBI directly counteracting the negative consequences.

Appendix B: Analysis of Evidence For and Against Basic Income as a Solution to Automation-Induced Job Loss

This document outlines the critical analysis of evidence regarding the efficacy of basic income (BI) or universal basic income (UBI) as a response to job displacement caused by automation and artificial intelligence (AI).

Arguments Supporting Basic Income

1. **Addressing Technological Unemployment:**

    - Evidence from sources like Subaveerapandiyan & Shimray (2024), Georgieff & Hyee (2022), and Wang & Lu (2025) indicates significant ongoing and projected job displacement due to AI and automation. While some studies (e.g., Wang & Lu, 2025; Autor, 2015 as cited by Wang & Lu) point to complementary effects and job transformation, the risk to certain sectors and skill levels remains high (Frey & Osborne, 2017, as cited by Wang & Lu, 2025).

    - BI is proposed as a mechanism to provide economic security when traditional employment becomes scarce or precarious for a large segment of the population.

2. **Poverty Alleviation and Improved Well-being:**

    - Systematic reviews like Rizvi et al. (2024) and the Stanford Basic Income Lab review (Hasdell, 2020) suggest that BI/GBI interventions can lead to improvements in poverty-related outcomes, particularly food security (Rizvi et al., 2024), subjective financial well-being, life satisfaction, and mental health (Rizvi et al., 2024).

    - The UChicago Urban Labs review (2024) also points to positive trends in financial well-being from US pilots.

- This suggests BI could mitigate the negative socio-economic consequences of job loss due to automation.

3. **Enhanced Worker Power and Entrepreneurship:**

    - A basic income floor could empower workers by providing greater bargaining power with employers and the option to refuse precarious or low-quality work.

    - Some literature suggests BI might foster entrepreneurship and risk-taking by providing a safety net, though direct evidence from large-scale UBI trials specifically linked to automation-driven job loss is still emerging.

    - Wang & Lu (2025) note AI's integration contributes to entrepreneurial opportunities, and BI could potentially support individuals pursuing these.

4. **Stimulating Economic Demand:**

    - By providing regular income to a broad population, BI could stabilize or increase aggregate demand, potentially counteracting deflationary pressures from automation-driven productivity gains that are not widely shared through wages.

5. **Simplification of Welfare Systems:**

    - Proponents argue UBI could simplify complex and often stigmatizing existing welfare programs, potentially reducing administrative costs (Rizvi et al., 2024, background section).

Arguments Against or Raising Concerns about Basic Income

1. **Labour Market Disincentives:**

    - A primary concern is that unconditional cash payments might reduce incentives to work, leading to labour shortages in certain sectors or an overall decline in

labour force participation. The evidence from BI experiments on labour supply is mixed and often context-dependent. Some studies show minimal impact, while others indicate slight reductions, particularly among secondary earners or those in education/caregiving roles.

- Hoynes & Rothstein (2019, cited in search results) review labour supply literature and express scepticism that ongoing UBI pilots will resolve these questions definitively.

2. **Economic Feasibility and Funding:**

   - The macroeconomic cost of implementing a truly universal and adequate basic income is substantial. Funding such a program would likely require significant tax reforms or reallocation of government spending, the political feasibility and economic impact of which are major points of debate.

3. **Inflationary Risks:**

   - A large injection of demand without a corresponding increase in productive capacity could lead to inflation, potentially eroding the real value of the basic income payments.

4. **Potential for Misuse of Funds:**

   - While often not borne out by evidence from pilot programs (which generally show funds are spent on necessities), a persistent concern among some critics is that unconditional cash might be used irresponsibly.

5. **Implementation Challenges:**

   - Defining the appropriate level of BI, universality, conditionality (if any), and integration with existing social safety nets presents complex design and implementation challenges.

- Rizvi et al. (2024) highlight the heterogeneity in outcomes and the difficulty in synthesizing results due to varied program designs and measurement instruments.

6. **Doesn't Address Non-Financial Aspects of Work:**

    - Critics argue that BI primarily addresses the income aspect of job loss but may not adequately compensate for the non-financial benefits of work, such as social interaction, sense of purpose, and identity.

7. **Risk of Political Capture or Inadequacy:**

    - The level of BI could become a political tool, potentially set too low to be meaningful or subject to erosion over time due to political pressures or inflation.

Conclusion of Initial Analysis

The evidence suggests that while basic income holds promise for addressing some negative consequences of automation-induced job displacement, particularly in terms of poverty reduction and improved well-being, significant concerns regarding labour supply effects, economic feasibility, and implementation complexities remain. The impact of BI is highly dependent on its design, the economic context, and the specific outcomes measured. A critical evaluation points towards a nuanced view rather than a simple 'yes' or 'no' answer to its suitability as *the* solution.

Further analysis is required on:

- The specific design features of BI that are most effective.
- The interaction between BI and other policy measures (e.g., reskilling, education reform).
- Long-term impacts and scalability of BI programs.

Appendix C: Common Themes, Debates, and Research Gaps

This document synthesizes common themes, ongoing debates, and identified research gaps from the literature on automation/AI-induced job displacement and basic income as a potential solution.

Common Themes

1. **Inevitability vs. Malleability of Technological Unemployment:**

    - A recurring theme is the tension between the view that widespread technological unemployment is an inevitable consequence of AI and automation (e.g., concerns raised by Frey & Osborne, 2017) and the perspective that policy interventions, skill adaptation, and new job creation can mitigate these effects (e.g., Autor, 2015; complementary effects noted by Wang & Lu, 2025).

    - The bibliometric review by Subaveerapandiyan & Shimray (2024) shows a growing body of research attempting to understand this dynamic.

2. **Heterogeneity of Impact:**

    - Both the automation literature (e.g., Wang & Lu, 2025, on sectoral and demographic differences) and the basic income literature (e.g., Rizvi et al., 2024, on varied outcomes across GBI types and populations) emphasize that impacts are not uniform. Certain industries, skill levels, and demographic groups are more vulnerable to job displacement, and different BI designs have varying effects on diverse populations.

3. **Multidimensionality of Well-being and Poverty:**

    - Basic income studies increasingly recognize that poverty and well-being are multidimensional. Beyond income, factors like mental health, physical health,

social inclusion, education, and agency are crucial outcomes (Rizvi et al., 2024; Stanford review, Hasdell, 2020).

- The loss of a job due to automation also has non-monetary impacts (loss of purpose, social connection) that BI alone may not address.

4. **Importance of Policy Design in Basic Income:**

    - The effectiveness of BI is heavily dependent on its specific design: the level of payment, universality vs. targeting, conditionality, funding mechanisms, and integration with existing social safety nets (Rizvi et al., 2024; UChicago review, 2024).

    - There is no one-size-fits-all BI model.

5. **Focus on Transitions and Adaptation:**

    - A theme emerging is the need for policies that support workers through transitions, including reskilling, upskilling, and lifelong learning, alongside or as an alternative to BI (OECD AI Principles, as cited by Subaveerapandiyan & Shimray, 2024).

Ongoing Debates

1. **Labour Supply Response to Basic Income:**

    - This remains one of the most significant debates. Concerns persist about potential negative impacts on work incentives, while proponents argue effects are often small, concentrated in specific groups (e.g., caregivers, students), or reflect a positive shift away from undesirable work. The evidence is mixed and context-specific (Rizvi et al., 2024; Hoynes & Rothstein, 2019).

2. **Economic Feasibility and Macroeconomic Impacts of UBI:**

    - The large-scale fiscal cost of UBI and the optimal methods for funding it (e.g., carbon tax, financial transaction tax, changes to income/corporate taxes) are heavily debated. Potential inflationary effects or impacts on investment and economic growth are also key concerns.

3. **UBI as a Primary Solution vs. Part of a Broader Policy Mix:**

    - Is UBI *the* answer to technological unemployment, or should it be one component of a broader strategy that includes active labour market policies, education reform, and industrial policy?

4. **Universality vs. Targeting:**

    - While UBI implies universality, debates continue about whether more targeted approaches (e.g., guaranteed minimum income, negative income tax) might be more efficient or politically feasible, though potentially more complex or stigmatizing.

5. **The Role of Work in Society:**

    - Underlying many debates is a philosophical discussion about the role of work beyond income – its contribution to identity, social cohesion, and purpose. If automation significantly reduces the need for human labour, how will societies adapt, and can BI help manage this transition?

Research Gaps

1. **Long-Term Effects of Basic Income:**

    - Most BI experiments have been relatively short-term (1-5 years, as noted by Rizvi et al., 2024). There is a need for more research on the long-term societal, economic, and individual impacts of sustained basic income.

2. **Impact of BI in the Specific Context of Automation-Driven Job Loss:**

    - While BI is proposed as a solution, few, if any, large-scale experiments have been explicitly designed to test its efficacy in communities already experiencing significant job displacement directly attributable to AI and automation. Most studies focus on general poverty alleviation.

3. **Scalability and Political Economy of UBI:**

    - More research is needed on the practical challenges of scaling up BI to a national level, including administrative systems, political consensus-building, and managing public perceptions and expectations.

4. **Interaction Effects with Other Policies:**

    - How does BI interact with other policies aimed at mitigating technological unemployment, such as investments in education, reskilling programs, or industrial strategy? Are there synergistic or conflicting effects?

5. **Standardized Metrics and Cross-Experiment Comparability:**

    - Rizvi et al. (2024) highlight the difficulty in synthesizing results due to heterogeneous outcomes and measurement tools. A need exists for more standardized approaches to allow for better cross-study comparisons.

6. **Impact on Innovation and Entrepreneurship:**

    - While some suggest BI could foster risk-taking, more robust empirical evidence is needed on its actual impact on innovation, new business formation, and the types of entrepreneurial activities pursued, especially in an automation context.

7. **Differential Impacts Across Diverse Economic and Social Contexts:**

    - Much of the prominent research (e.g., Rizvi et al., 2024) focuses on high-income countries. More research is needed on the feasibility and impact of BI in low- and middle-income countries facing similar technological pressures.